Fairgame

The Decentralized Autonomous Propfirm

An accessible introduction to a self-balancing,
verifiable proprietary-trading protocol on Sonic

Whitepaper · Version 1.0 · October 2026

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Companion to the *Fairgame Yellow Paper*, Version 1.0

Abstract. This whitepaper describes Fairgame, a decentralized autonomous proprietary-trading firm deployed on Sonic, an EVM-compatible layer-1 blockchain chosen for its fast finality and low fees. Traders buy evaluations in stablecoins, trade a non-transferable unit of account, Sonic Virtual USD (svUSD), against oracle prices in a fully on-chain simulator, and, once they pass, operate funded svUSD accounts whose profits are paid out in USDC from a transparent reserve. Conventional prop firms stay solvent through opaque pricing, discretionary payout denial and identity checks, and their customers can verify neither payout decisions nor reserves; firms marketed as crypto or decentralized prop firms often change only the payout rail. Fairgame makes its rules, account states, reserve balances, obligations and payouts, and every price it uses, verifiable on-chain at every block; since none of the conventional levers is available to a permissionless protocol, it derives its prices from first principles instead. The Fairgame Yellow Paper proves that, under a fair-execution discipline and explicit assumptions on oracle prices and keepers, every trader without an informational edge faces a supermartingale, up to one stated gap for conditional orders. From this, we derive closed-form upper bounds on the probability of passing an evaluation and on the lifetime payout of any funded account. These bounds define a *farm-proof price* at or above which no strategy without an informational edge has positive expected value, a property the Yellow Paper proves extends to arbitrary coalitions of Sybil accounts, together with a liability measure that can be evaluated at every block. On this valuation layer sits a feedback controller that adjusts prices, the menu of offered rule sets and funded capacity from realized cash flows and live coverage; we summarize the stability region of its price loop and the adverse-selection condition under which price stops being an effective lever, both derived in the Yellow Paper. The design further includes a priority fee waterfall, a stress-coverage rule that bounds exposure to correlated winners, solvent-by-construction tournaments for bootstrapping, optional proof-of-personhood discounts with bounded exposure, a verifiable trader passport, and a copy-capital vault that turns statistically skilled traders from a liability into a source of revenue.

All parameters in this document are proposals subject to calibration. Nothing in this document is an offer of any product, investment advice or legal advice.

Contents

1 Executive summary **1**

2 The problem with conventional prop firms **1**

2.1 How the evaluation business works 1

2.2 Levers customers cannot see 1

2.3 "Crypto" prop firms often change only the payout rail 2

3 The Fairgame solution **2**

4 How Fairgame works **3**

4.1 Participants 3

4.2 The account lifecycle 3

4.3 svUSD: the simulated unit of account 3

4.4 Rule sets 4

4.5 Fair execution: separating the decision from the price 4

4.6 Pass, breach and payouts 5

5 Why Sonic **5**

6 Farm-proof pricing **6**

6.1 No free edge 6

6.2 Three numbers that bound every strategy 6

6.3 Worked examples 6

6.4 Sybil resistance by pricing 7

6.5 Which rules matter for price 8

7 Solvency and fairness guarantees **8**

7.1 A liability anyone can read 8

7.2 The skill ratio: a built-in leak detector 8

7.3 Stress coverage: bounding correlated winners 8

7.4 Protocol invariants 9

8 Treasury and fee waterfall **9**

8.1 Five vaults 9

8.2 The priority waterfall 9

8.3 Network revenue 9

9 The self-balancing controller **10**

10 Growth mechanisms **11**

10.1 Tournaments: the launch product 11

10.2 Optional proof of personhood and discounts 11

10.3 The trader passport 11

10.4 The copy-capital vault (later phase) 12

11 Governance **12**

12 Security, risks and open questions **13**

13 Regulatory considerations **14**

14 Roadmap **14**

15 Conclusion **15**

Disclaimer. All parameters in this document are proposals subject to calibration. Nothing in this document is an offer of any product, investment advice or legal advice. This document describes a protocol design for informational purposes. Statements about the protocol's guarantees hold only under the assumptions stated in the Fairgame Yellow Paper, which remains the authoritative specification. Descriptions of third-party systems (Sonic, Pyth, Chainlink, Circle and others) reflect the sources cited in the Yellow Paper as of October 2026 and may change.

1 Executive summary

Fairgame is a decentralized autonomous proprietary-trading firm ("prop firm") designed to run entirely on-chain on Sonic. Traders buy evaluations in stablecoins, trade a simulated account against live oracle prices, and, once they pass, receive a funded account whose profits are paid out in USDC from a public reserve. Every rule, account balance, reserve balance, liability and payout is meant to be verifiable on-chain at every block.

Fairgame in six points

- **Fair execution.** Orders are committed first and filled at the first oracle price published after a short delay, so no one can trade on stale prices or pick a favorable fill.
- **Farm-proof pricing.** Under fair execution, a trader without a real informational edge faces at best a fair game. That caps what any such strategy, or any network of fake ("Sybil") accounts, can expect to extract, and evaluations are never sold below that cap.
- **No discretion.** Payouts follow a published formula on on-chain state, are released within a bounded window and can be challenged only with objective evidence, never over trading style.
- **Public solvency.** The protocol's expected liability, its reserve and its coverage ratio can be read at every block. A stress-coverage rule limits exposure to many traders winning at once.
- **Self-balancing economics.** A feedback controller adjusts prices, the menu of rule sets on sale and funded capacity from realized cash flows, within hard bounds and with a published stability analysis.
- **Built for growth.** Tournaments that are solvent by construction bootstrap the reserve. Optional proof-of-personhood discounts, a verifiable trader passport and a planned copy-capital vault turn skilled traders from a cost into a source of revenue.

This whitepaper is the accessible companion to the *Fairgame Yellow Paper* (v1.0, October 2026), which states the protocol formally and contains the proofs. Where this document simplifies, the Yellow Paper is authoritative. All parameters mentioned here are **proposals subject to calibration**.

2 The problem with conventional prop firms

2.1 How the evaluation business works

A retail prop firm sells evaluations. A trader pays a fee, receives a simulated account of a nominal size, and must reach a profit target before breaching a set of loss limits. Traders who pass get a funded account (at most firms also simulated) and are paid a share of the profits it records, typically 70–90%. Economically, the fee is a premium for a claim on the trader's performance, and the firm is solvent when premiums exceed expected payouts plus operating costs.

2.2 Levers customers cannot see

Conventional firms keep that inequality with three levers that are largely invisible to their customers:

(i) **Behavioral under-performance.** Evaluations are typically priced below what a disciplined random strategy would expect to earn. The model works because most customers

do worse than random once costs, over-trading and time pressure are counted.

(ii) **Discretion.** Payouts can be denied after the fact under loosely defined rules on "gambling", "consistency" or "prohibited strategies", whose scope the firm decides for itself.

(iii) **Identity.** Know-your-customer checks and linking accounts through shared devices and payment details make it hard to run many accounts, and with them to systematically harvest under-priced evaluations.

The model is also **opaque**. Published terms can make every payout subject to the firm's final approval. Belgium's financial regulator (FSMA) has warned that, because the firm alone sets the conditions of use, consumers find it hard to tell whether they are owed anything, and if so how much. Italy's securities regulator (Consob) has received complaints that promised profit shares were not paid. Customers generally cannot see the reserves they are paid from, or tell whether the firm will keep operating: in February 2024, several firms suspended trading after losing access to their platform, one deferring all payouts.

2.3 "Crypto" prop firms often change only the payout rail

Firms marketed as crypto or decentralized prop firms often change only how they pay: stablecoin payouts, while evaluation, account state and rule enforcement stay off-chain and reserves go unpublished. Newer entrants may publish payout contracts or pay out in USDC on-chain, but in the designs reviewed for the Yellow Paper the trading engine typically still runs off-chain or under operator control, with evaluations sold from fixed price lists.

Why simply copying the model on-chain fails
None of the three levers survives in a permissionless setting. Bots run disciplined random strategies at scale, discretionary code is a trusted party in disguise, and wallets are free. A decentralized prop firm that copies conventional pricing is *exploitable by construction*: multi-account hedging (go long on one evaluation, short on another, collect from whichever passes) would drain it.

3 The Fairgame solution

Fairgame puts the trading engine itself on-chain, derives prices from a valuation theory that holds against every strategy without an informational edge, and replaces fixed price lists with feedback control. Every price, rule and capacity limit is either derived from that theory or set by an on-chain controller within hard bounds. **Nothing is discretionary.**

The name refers to the mathematical notion of a *fair game* (a martingale). The protocol's central result is that, under fair execution, trading on Fairgame is at best a fair game for a trader without an edge. Fairgame exists to find and pay real skill, not to profit from customers' mistakes or to deny payouts.

Design goals. *Verifiability:* every state change can be reproduced from on-chain data and signed oracle reports. *Solvency without discretion:* payouts are a deterministic function of state, and solvency comes from pricing and invariants, not from a right to refuse payment. *Permissionless access:* identity is optional, and the protocol is designed to stay safe against any number of anonymous accounts. *Self-balancing economics:* prices, the rule menu and funded capacity respond automatically to cash flows and live liabilities. *Trader fairness:* rules are frozen at purchase, published in full and enforced identically for everyone. *Sustainability:* revenue flows through a priority waterfall into a payout reserve, an insurance fund, a sustainability fund and development,

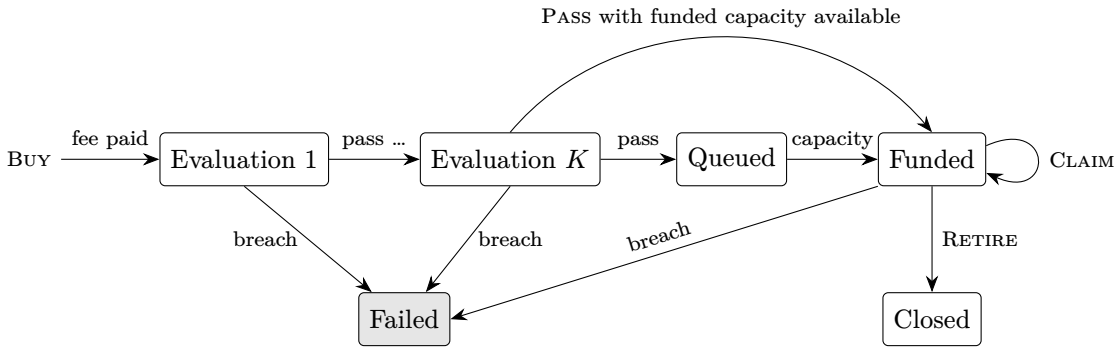


Figure 1: Account lifecycle. Rules are frozen at BUY. Anyone can prove a breach from signed oracle prices. A passing account enters the funding queue only when funded capacity is exhausted.

owned and governed by the community. *Sonic-native:* fast finality, low fees, fee monetization, gas sponsorship and native USDC are built into the design.

Design principles. (1) Rules are immutable per account; parameters may change for new accounts only. (2) Trades are against the oracle: no counterparty, order book or internal matching, hence no wash trading, self-matching or liquidity games. (3) Every liability is priced at its zero-skill value (Section 6). (4) No discretion: payouts are a deterministic function of on-chain state. (5) Feedback over prediction: aggregate signals drive parameters within hard bounds (Section 9).

4 How Fairgame works

4.1 Participants

- **Traders** buy evaluations, trade svUSD accounts and claim payouts.
- **Keepers and watchers** settle orders, execute limit and stop orders, prove rule breaches and trigger the daily controller update. All of these roles are permissionless and paid per action.
- **Oracles** (Pyth as primary, Chainlink Data Streams as secondary) publish signed price reports that the engine verifies on-chain.
- **Governance** maintains rule sets, market listings and controller parameters within immutable bounds; **liquidity providers** (later phase) fund the copy-capital vault.

4.2 The account lifecycle

An account is minted when a trader buys an evaluation, and its rule set is frozen at that moment. It moves through one or two evaluation phases by meeting their profit targets, and then becomes *funded*. If funded capacity is temporarily exhausted, it waits in a first-in, first-out funding queue instead. A funded account records claims until it breaches a loss limit or its owner retires it. Each account is a non-transferable (soulbound) NFT locked to its owner.

4.3 svUSD: the simulated unit of account

Simulated balances, positions and profits are recorded in **Sonic Virtual USD (svUSD)**. svUSD is not a token: it exists only as a field in an account record, and it has three properties.

Non-transferable. Nothing moves svUSD between accounts or addresses, and there is no token contract, which removes a whole class of attacks.

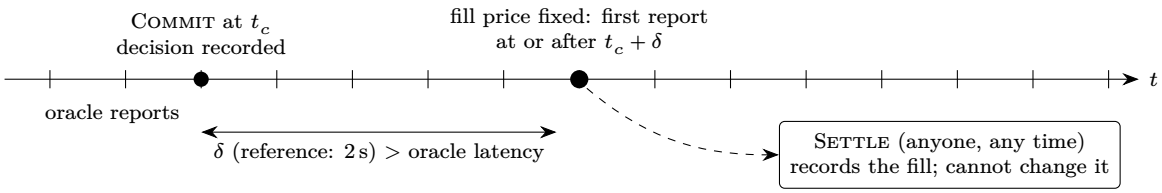


Figure 2: Two-step execution of a market order. The outcome is fixed at commit time, up to the oracle's future output, so no party (including a trader who also runs a keeper) holds an option after committing.

Non-redeemable. svUSD is never exchanged for any asset. USDC payouts are performance rewards computed from recorded profits, not redemptions.

Engine-only. Balances change only through engine transitions: fills, fees, claims and phase resets.

svUSD and USDC share a numerical scale so that rules read naturally (an account with a balance of 25,000 is "a $25k account"), but the only bridge between them is the payout function.

4.4 Rule sets

Each evaluation is sold under a **rule set** from a governance-approved catalog. It fixes the account size, one or two phases with their profit targets and maximum losses, daily loss limits, the funded size, the profit split, a leverage cap, the number of open markets, a minimum claim, an optional first-claim fee refund and the tradable markets.

Some rules are *deliberately absent*: minimum trading days, consistency rules and bans on particular strategies. They are not needed for solvency, and they are the rules traders most often experience as arbitrary.

4.5 Fair execution: separating the decision from the price

Execution is where a simulator leaks value: any fill better than the market hands out a free edge, and the entire pricing theory depends on there being none. Fairgame therefore splits a market order into two steps. The trader first COMMITS the order on-chain. The fill price is then the price of the *first* oracle report published at or after the commit time plus a short commit delay δ, which is longer than the oracle's latency. Anyone can SETTLE the order at any later time, but settlement only records a fill that was already determined.

- **Market orders cannot be canceled**, and an account holds at most one pending market order. Limit and stop orders execute only on a fresh, admissible report that meets the trigger, and cancellations take effect only after the commit delay.
- **Two oracles must agree.** A Pyth price is used only if a near-simultaneous Chainlink report agrees within a tolerance and its confidence interval is narrow enough.
- **Costs are never credited.** Fills pay a spread (base, plus oracle uncertainty, plus price impact) and a trading fee. The crowded side of a market pays a one-sided crowding fee that, unlike funding on perpetual venues, is never credited to the other side, since a credit would let bots farm the minority side.
- **Risk checks** admit a risk-increasing fill only within the leverage cap, the open-market limit and, for funded accounts, the stress-coverage rule (Section 7.3). Risk-reducing fills are always admitted.

4.6 Pass, breach and payouts

Breach. An account fails when its equity, measured on a synchronized set of admissible prices, falls to or below its static or daily loss floor. Anyone may prove a breach and earns a bounty from the account's escrow. Requiring synchronized prices stops a prover from combining prices from different moments to fabricate a loss on a hedged book.

Pass. A flat account (no positions or pending orders) that has reached its profit target moves to the next phase, or to funded status if funded capacity is available.

Claims. A flat funded account with profit G above the minimum claim receives αG **in USDC** from the payout reserve, where α is the profit split. Its balance then resets to the funded size and the loss floor stays where it was, so every claim cycle starts the same distance from failure. Small claims are paid instantly. Larger claims are released after a challenge window (hard bound: at most 72 hours), during which they can be challenged *only with objective evidence*: an oracle incident affecting one of the account's fills, or a proof that a protocol invariant was violated. **Trading style can never be challenged.**

5 Why Sonic

Every commit, settlement, conditional execution and breach proof is a transaction, so the chain decides how fast actions take effect and what they cost. Speed is the main reason Fairgame is built on Sonic, an EVM-compatible proof-of-stake layer-1 with fast, deterministic finality.

Speed and finality. Sonic's asynchronous BFT consensus gives deterministic rather than probabilistic finality. Sonic Labs states that transactions are irreversible in under a second, and reports an average of 0.58 s per block and a peak of 1,465 transactions per second during the market-wide liquidations that began on 10 October 2025. The design assumes only finality within one to two seconds.

What speed buys. Fairness does not come from speed; speed sets what fairness *costs*. Fast inclusion means low latency between decision and fill, short keeper windows and therefore narrow spreads (a window ten times longer would need a spread floor about 3.2 times wider), and breach proofs that land within seconds.

Low, predictable fees. Sonic's base fee has a floor of 50 gwei and, according to Sonic Labs, held at that floor through the October 2025 liquidations. At the floor, a one-million-gas transaction costs 0.05 S, so keeper rewards and gas sponsorship stay cheap even in volatile markets.

Native rails and tooling. Native USDC with Circle's CCTP V2 (since May 2025) for payouts; Pyth and Chainlink Data Streams both deployed; protocol-level gas sponsorship, so traders never need to hold S; ERC-4337 and EIP-7702; and Fee Monetization, which currently returns 90% of the fees for gas used in an application's contracts.

Trade-offs. Sonic is safe only while validators holding less than a third of the stake are faulty. A chain halt delays every transition and acts like a price gap for limit and stop orders. Fee Monetization may be restructured, so it is treated as upside rather than as a revenue line the design depends on. Sonic's mainnet has run only since December 2024.

6 Farm-proof pricing

Pricing is the core of Fairgame. It rests on a single benchmark: the **zero-skill trader**, whose orders use only public information. That covers every technical, momentum and martingale-betting system based on public data, run on any number of wallets in any combination. *Skill* means an informational or predictive edge the market has not yet priced.

6.1 No free edge

Under fair execution, and explicit assumptions (oracle prices behave like an efficient-market benchmark over an account's horizon, oracle latency is bounded, keepers are live), a zero-skill trader's equity is at best a fair game: costs are non-negative and no fill can beat the market. This is the formal statement that *the simulator does not leak*. The proof pins down which design choices carry the weight: a commit delay above oracle latency, first-report market fills, no credits, and a base spread that covers the small value of a keeper choosing among fresh reports (about 1.2 basis points for 60% annual volatility and a 2-second window). One technical gap remains for limit and stop orders (Section 12).

6.2 Three numbers that bound every strategy

Once trading is at best a fair game, a classical result (the gambler's ruin) gives closed-form ceilings that hold for *any* zero-skill strategy, with any position sizing, number of trades or stopping rule. For a phase with profit target T and maximum loss D (both as fractions of the account size), the pass probability is at most

$$\bar{p} \;=\; \frac{D}{T+D}. \tag{1}$$

To pass, an account must travel a distance T before it travels a distance D the other way; without an edge, the odds are no better than the ratio of the distances. (In the full formula, D is enlarged slightly by the overshoot beyond the floor before a breach is caught, which is negligible when limits are checked every few seconds.) For two phases, the probabilities multiply.

A funded account of size S_f with maximum loss D_f works like a reserve of $D_f S_f$ virtual dollars that a zero-skill trader can at best convert into claims one for one before it runs out; the profit split α converts those into USDC. The expected lifetime payout is therefore at most

$$V_f \;=\; \alpha\, D_f\, S_f. \tag{2}$$

How often or when the trader claims changes the timing of payouts, but not their expected total. Combining the two gives the **zero-skill value** $V^\star = \bar{p}\, V_f$ of an evaluation, and the **farm-proof price**

$$F^\star \;=\; \frac{V^\star + c_{\text{op}}}{1 - r_{\text{ref}} - \rho\,\bar{p}}, \tag{3}$$

where c_{op} is the per-account operating cost, r_{ref} the referral share and ρ the fraction of the fee refunded on the first claim. Referrals sit in the denominator because self-referral is free in a permissionless system, so a commission is economically a discount. Anonymous buyers are never charged less than $F^\star$: the price multiplier is hard-bounded at ≥ 1.

6.3 Worked examples

Take a one-phase \$25,000 evaluation with a 10% target, a 5% maximum loss in both the evaluation and the funded stage, and an 80% split, ignoring overshoot, costs, referrals and refunds. The pass bound is $5/15 = 1/3$, the funded value is $0.8 \times 5\% \times \$25{,}000 = \$1{,}000$, and the farm-proof

Table 1: Zero-skill values of representative rule sets (split 80%, evaluation size = funded size; overshoot, costs, referrals and refunds ignored, so $F^\star = V^\star$). Source: Yellow Paper, Appendix B.

Rule set	Size	Targets	Max loss	$\bar{p}$	V_f	$F^\star$
Tight	\$10,000	8%	3%	0.2727	\$240	\$65.45
Standard tight	\$25,000	10%	3%	0.2308	\$600	\$138.46
Standard	\$25,000	10%	5%	0.3333	\$1,000	\$333.33
Wide	\$50,000	10%	6%	0.3750	\$2,400	\$900.00
Two-phase	\$100,000	8%, 5%	5%	0.1923	\$4,000	\$769.23
Two-phase wide	\$100,000	10%, 5%	10%	0.3333	\$8,000	\$2,666.67

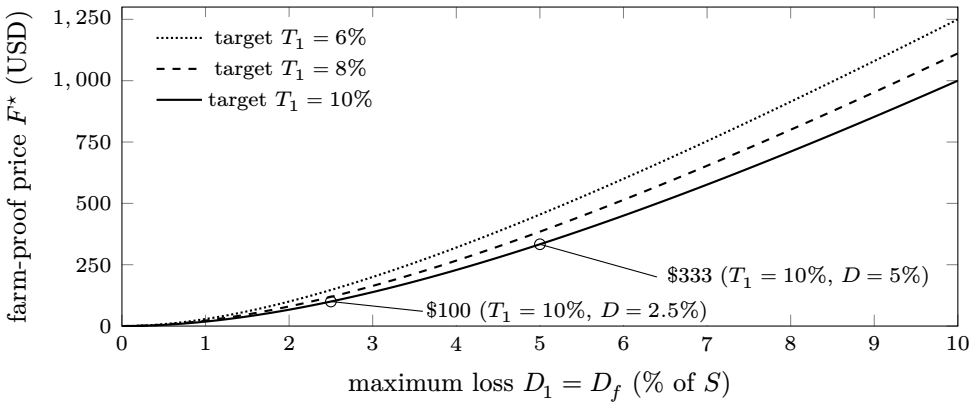


Figure 3: Farm-proof price of a one-phase \$25,000 evaluation with an 80% split as a function of the maximum loss, ignoring overshoot, costs, referrals and refunds: $F^\star = \alpha S D^2/(T_1 + D)$.

price is **\$333.33**. Halving both loss limits to 2.5% lowers it to **\$100**. Table 1 reproduces the representative rule sets of the Yellow Paper.

The last row uses loss limits typical of conventional two-phase evaluations. Its farm-proof price is several times what evaluations with similar parameters commonly sell for, a measure of how far conventional pricing relies on customers doing much worse than random. Under farm-proof pricing, affordability comes from **tighter rules** instead. The price is convex in the loss limit (Figure 3), so tightening the limit is the cheapest way to offer affordable evaluations. With a full fee refund on the first claim, the standard rule set would need a price of \$500.

6.4 Sybil resistance by pricing

The best-known attack on prop firms is multi-account hedging: buy two evaluations, go long on one and short on the other, and collect from whichever passes. The Yellow Paper proves that a coalition controlling *any* number of accounts, bought at any times under any rule sets and traded with any jointly chosen zero-skill strategy, has non-positive expected profit whenever every evaluation is priced at or above its farm-proof price. This holds even when the coalition collects its own referral commissions, refunds and keeper rewards.

Key insight: Sybil resistance by pricing

Correlation between accounts changes the *variance* of a coalition's result, not its *expectation*. A hedged pair turns a lottery into a near-certain result with the same mean. Multi-account farming pays only when evaluations are under-priced, so a protocol that prices at or above $F^\star$ needs no identity layer to stay solvent. Identity becomes a tool for offering *discounts* safely, not a precondition for survival.

6.5 Which rules matter for price

Daily loss limits, minimum trading days, consistency rules, position caps, trading costs and claim minimums can only *lower* what a zero-skill strategy extracts, so the price does not depend on them. By contrast, anything that raises the zero-skill value must be priced in: larger funded sizes (scaling), refunds, wider loss limits, and anything that lets losses overshoot the floor, such as checking limits only at the end of the day or monitoring less often.

7 Solvency and fairness guarantees

7.1 A liability anyone can read

The same bounds apply to every live account at any moment, so the protocol's total expected liability can be computed and published continuously. A **funded** account's expected remaining payout is at most α times its equity minus the level at which it would fail. An **evaluation** account carries its current pass bound times the funded value, so liability is recognized gradually as an account approaches its target and passing causes no jump.

For example, a funded standard \$25,000 account at equity \$26,000 carries a zero-skill liability of $0.8 \times (\$26{,}000 - \$23{,}750) = \$1{,}800$; at \$24,000 it carries \$200. A conventional firm can estimate this figure internally; Fairgame publishes it. A public firm-health page shows the reserve, liabilities, coverage, loss and skill ratios, stress coverage, the queue, the menu and prices; before buying, a trader sees the farm-proof price and zero-skill pass bound of every rule set.

7.2 The skill ratio: a built-in leak detector

For each rule set and market, the protocol tracks the **skill ratio**: realized payouts on closed accounts divided by their zero-skill value at purchase. A population without skill should come out at or below 1, apart from sampling noise. A persistently higher ratio means either that genuinely skilled traders are present or that the simulator leaks somewhere. Both deserve attention, and the second deserves it urgently. A conservative upper estimate of this ratio scales the published liability upward.

7.3 Stress coverage: bounding correlated winners

Pricing controls the *expected* liability, but markets trend. Many funded accounts long the same asset in a rally create a tail liability that pricing does not address, and Fairgame bounds it directly. The **stress liability** L_{st} is what funded accounts could claim right now, plus what they could gain if every market moved by a stress amount (reference proposal: 15% for BTC and ETH, 25% for S and SOL) on their gross open interest. The **stress-coverage invariant** requires

$$V_{\text{res}} \;\geq\; (1+\beta)\, L_{\text{st}} \tag{4}$$

at every risk-increasing funded fill, every admission to funded status and every reserve sweep, where V_{res} is the payout reserve net of pending claims and β a buffer. Transitions that would violate it are rejected. While prices stay within the stress moves of their values at a time when the invariant held, all USDC that funded accounts can claim is provably covered by the reserve. In effect this is a dynamic open-interest cap that tightens automatically as the reserve shrinks or prices rise, and the one-sided crowding fee prices the herding that uses up the headroom. One limitation is stated openly: price moves can erode stress coverage between checks, and actively restoring it is an open design question.

7.4 Protocol invariants

Alongside stress coverage, the Yellow Paper specifies machine-checkable invariants for property-based testing and formal verification: **accounting** (protocol USDC equals vaults plus escrows plus unsettled tournament buy-ins); **backed claims** (pending claims never exceed the reserve); **rule immutability**; **non-transferability** of balances; **fair timing** of every fill; **non-negative costs**; **leverage** limits after every risk-increasing fill; **bounds** on all parameters (anonymous multiplier ≥ 1); and **deterministic payouts** released within the challenge window.

8 Treasury and fee waterfall

8.1 Five vaults

Fairgame holds USDC in five on-chain vaults that anyone can audit, plus a small escrow per account that pays keeper rewards and breach bounties:

- the **payout reserve**, the only source of trader payouts; pending claims are held inside it and excluded from every coverage calculation;
- the **insurance fund**, a capped backstop for events outside the model (oracle incidents, extreme price moves, contract defects), targeted at a multiple of the stress liability;
- the **sustainability fund**, for recurring costs (oracle updates, keeper subsidies, gas sponsorship) and the operating runway, held in conservative instruments; it also receives Fee Monetization income;
- the **development fund**, for contributors, audits and bug bounties;
- an optional **burn allocation**, which buys back and burns S, Sonic's native token, in line with Sonic Labs' stated expectation that applications receiving its grants return value to the protocol layer. This is not a Fairgame token.

8.2 The priority waterfall

Every evaluation fee is split in a fixed order of priority:

1. **Referral:** the referral share, if a referrer is named.
2. **Escrow:** a fixed amount into the account's own escrow for keeper rewards and bounties.
3. **Payout reserve:** always at least a minimum share of the net fee; while the reserve is below its coverage target, whatever is needed to close the gap, up to the whole net fee.
4. **Insurance fund:** a share, while the fund is below its target.
5. **Remainder:** split among sustainability, development and burn in fixed proportions.

Sustainability, development and burn therefore receive only what is left after the reserve and the insurance fund, and because the reserve always receives a minimum share it keeps growing with the book even when coverage is met. When coverage is well above an upper band (reference proposal: 2.5× the adjusted liability, against a target of 1.5×), a limited fraction of the excess may be swept to sustainability and burn each epoch, and only as far as the stress invariant allows.

8.3 Network revenue

Engine transactions are frequent, so Fairgame generates significant network fees. Under Sonic's Fee Monetization, the protocol currently receives 90% of the fees for gas used in its registered contracts, paid in S into the sustainability fund. All solvency and controller results hold with this revenue set to zero.

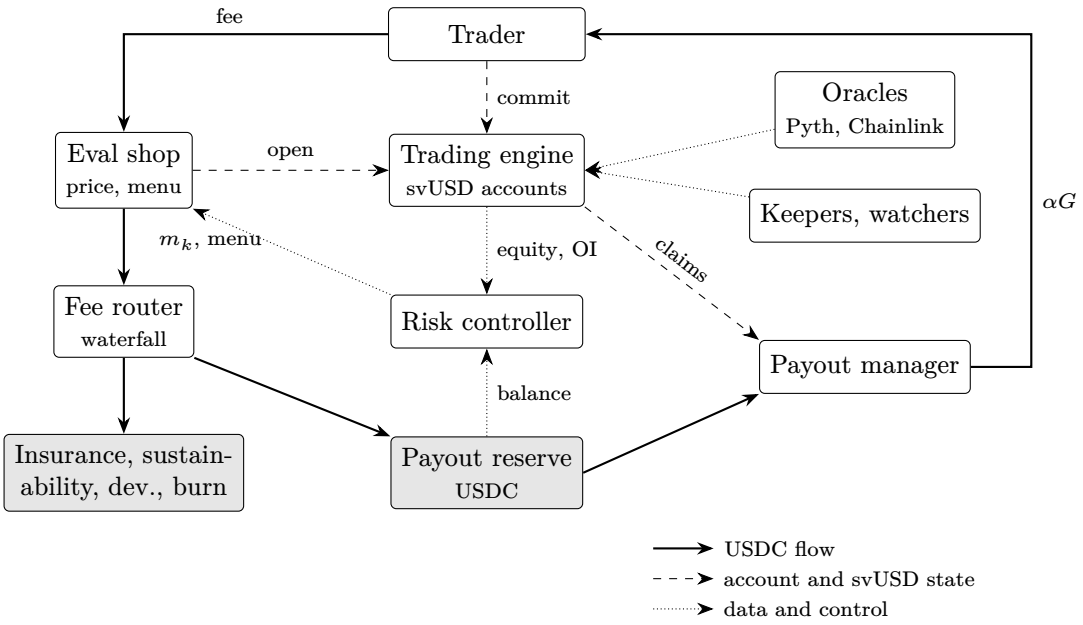


Figure 4: Value and information flows. Fees enter through the eval shop at the price set by the risk controller and are split by the fee router. Funded accounts turn recorded profits into claims, and the payout manager pays the profit share αG in USDC from the payout reserve. Extensions (tournaments, identity, copy-capital vault) are omitted.

9 The self-balancing controller

Farm-proof pricing is a *floor* set by the zero-skill benchmark. Real populations differ from it: most traders under-perform it, some genuinely out-perform it, and markets trend. A feedback controller closes the loop between realized cash flows and the parameters that create liabilities. It runs once per epoch (one day in the reference proposal) in a transaction that anyone can trigger.

Signals and levers. The controller observes *coverage* (reserve divided by adjusted liability), the *loss ratio* (smoothed claims paid divided by net fee revenue), the *skill ratio*, *stress coverage* and the *funding-queue length*. It acts through the *price multiplier* applied to the farm-proof price (hard floor: 1), the *menu* of rule sets on sale, *funded capacity* and, implicitly, open-interest headroom.

Price law. Prices equal the farm-proof price times a multiplier $m \geq 1$. Each epoch, the logarithm of the multiplier moves toward a target loss ratio, rises when coverage falls short of target or when the funding queue grows, and falls when coverage exceeds its upper band. Changes are rate-limited (hard bound: at most 10% per epoch; reference proposal: at most 5% per day), so traders face predictable prices.

Stability, stated in advance. The Yellow Paper derives the exact range of controller gains within which the price loop converges smoothly, converges with oscillation, or becomes unstable, and shows how lags between sales and claims shrink that range. The proposed reference gain sits in the smooth regime and tolerates a lag of up to two epochs. In the typical retail case prices rest at the farm-proof floor, and they rise above it only when realized payouts require it.

Adverse selection. Raising prices changes who buys. If casual traders leave first, the remaining population is more skilled and higher prices may not lower the loss ratio (the "lemons spiral" of

insurance markets). The controller estimates this effect continuously. When it crosses a threshold, the controller stops raising prices and switches to *quantity controls*: it tightens the menu, which cuts new liabilities directly.

The dynamic rule menu. Rule sets with the same zero-skill value are economically interchangeable for the protocol, and tighter rules make evaluations cheaper. When cash out outpaces cash in, coverage falls and the most generous rule sets (large accounts, wide loss limits) leave the menu, while tight rule sets stay on sale at unchanged prices. The rules of existing accounts never change.

Capacity and the funding queue. Passing creates no new expected liability, since it was recognized as the evaluation approached its target. What funding creates is the ability to claim USDC immediately, that is, stress exposure. A passing account is funded only when coverage is at target and the reserve has stress headroom for it to use its full leverage. Otherwise it joins a public first-in, first-out queue and is admitted automatically once capacity frees up. A long queue signals scarce capacity and pushes prices up.

10 Growth mechanisms

10.1 Tournaments: the launch product

Tournaments generate trading activity, a leaderboard and a first funded cohort before open-ended evaluations exist, and they **cannot create unfunded liabilities**. Entrants pay a fixed buy-in and trade an svUSD account under a set rule set over a fixed window; a prize share is distributed by rank, and the top entrants receive funded accounts. A simple condition, checked at the minimum number of entrants, guarantees that prizes, referrals, escrow and the full zero-skill value of every awarded funded account are paid out of buy-ins; the rest flows through the waterfall. Because prize money is fixed, multiple entries are a fairness question (handled by optional entry caps), not a solvency one.

Rather than raw return, which rewards lucky all-in bets, tournaments score **return per unit of maximum drawdown**, disqualify entrants who breach the loss limits and require a minimum number of trading days. Open evaluations launch once the reserve built from tournaments covers the planned initial menu.

10.2 Optional proof of personhood and discounts

Identity is never required. A trader may attach a unique-human attestation to an address, for example a zero-knowledge proof of personhood or of a government credential. This unlocks a **verified price** below the anonymous price for a capped number of evaluations per epoch. Because both the discount floor and the cap are bounded, the most a single verified identity can extract in expectation is bounded too, and farming discounts with fake identities becomes unprofitable once a fraudulent attestation costs more than that bound. This lets Fairgame compete with conventional firms on price for the honest majority without reintroducing multi-account risk.

10.3 The trader passport

Each address accumulates a soulbound **passport** with the on-chain history of its accounts: evaluations bought, passed and failed; claim cycles and payouts; maximum drawdowns; and a commitment to every funded account's daily equity series. From it anyone can compute the trader's Sharpe ratio and its *probabilistic* and *deflated* versions, which correct for the fact that among thousands of accounts some look skilled by chance. Reputation unlocks larger verified discounts

(within the caps), **scaling** of the funded size after successful claim cycles (each increase priced as new liability and subject to capacity), **direct funding** without an evaluation for verified traders with a high deflated Sharpe ratio (priced with a pass probability of one), and copy-vault eligibility and governance weight.

10.4 The copy-capital vault (later phase)

Simulated payouts make skilled traders a cost: the better they trade, the more the reserve pays. The copy-capital vault reverses this. Liquidity providers deposit USDC into an ERC-4626 vault with epoch-based withdrawals (ERC-7540), so that depositors cannot exit ahead of visible losses. An executor mirrors the positions of selected funded traders on an external perpetual-futures venue. Eligibility requires a deflated Sharpe ratio above a threshold over a minimum track record, drawdowns within limits and a verified passport, and copy weights are capped per trader and per market.

Copying a trader without skill never improves the expected result and only costs execution. For a genuinely skilled trader, copying at a weight above the profit split makes the combined result of reserve and vault grow with that skill and turn positive. Because commits are public before the simulated fill, the executor can trade at roughly the same moment. A net-book hedging overlay can reduce, but not remove, the reserve's option-like exposure.

11 Governance

The guiding rule: **governance may tune the protocol but may not change a promise already made.** The rules of existing accounts, claims already earned and the hard bounds are beyond its reach.

Table 2: Parameter classes.

Class	Changed by	Examples
Immutable	nobody	svUSD properties; immutability of account rules; no discretionary denial of claims; the hard bounds; the protocol invariants
Slow	timelocked governance (at least 7 days)	rule-set catalog; market listings and oracle sources; waterfall weights; controller gains and targets within bounds; spreads and fees
Automatic	the epoch controller	price multiplier; menu; queue admissions; reserve sweeps
Emergency	guardian	pause new evaluations and risk-increasing fills for a bounded time; never pause claims of settled profits or move funds

Table 3: Hard bounds (proposed initial values).

Parameter	Bound	Rationale
Anonymous price multiplier	≥ 1	farm-proof pricing
Commit delay	$\geq$ 99.9th percentile of oracle latency $+ 1\,\mathrm{s}$	no trading on stale prices
Base half-spread	$\geq$ residual selection value	no favorable report selection
Profit split	$\leq 90\%$	bounds the funded value
Leverage cap	$\leq 10\times$	bounds overshoot
Stress buffer β	≥ 0.25	shock solvency
Coverage target	≥ 1.25	reserve adequacy
Price rate limit	$\leq 10\%$ per epoch	price predictability
Challenge window	$\leq 72\,\mathrm{h}$	timely payouts
Guardian pause	≤ 7 days	bounded emergency power

Progressive decentralization. At launch, slow-class changes are proposed by a multisig and executed through a timelock that anyone can monitor. In the target state, voting power is *earned, not bought*: each verified passport holder receives non-transferable weight equal to the square root of the USDC they have received in payouts, scaled by funded tenure. The square root dampens concentration and the tenure factor resists short-term capture; weight is bound to a verified identity so that splitting payouts across addresses does not pay. The guardian (initially a multisig, later elected) can pause new evaluations and risk-increasing fills for a bounded time, but cannot pause claims of settled profits, alter balances or move funds, and every pause is logged on-chain with a stated reason.

12 Security, risks and open questions

The Yellow Paper assumes a strong adversary: it controls any number of addresses, runs keepers, watches the mempool and sees external prices ahead of the oracle. It cannot break cryptography, forge both oracle networks at once, control a third of Sonic's stake or hold a governance majority.

Table 4: Selected threats and mitigations (condensed from the Yellow Paper's threat model).

Threat	Mitigation	Residual risk
Latency arbitrage	commit delay above oracle latency; first-report fills; market orders cannot be canceled	none under stated assumptions
Oracle manipulation or failure	liquid markets only; Pyth report admitted only with an agreeing Chainlink report; confidence and staleness bounds	one network: prices within tolerance, or delay; both: tail risk, insured
Sybil farming, hedged coalitions	farm-proof pricing; discounts only with attestation and caps	cost of fake identities
Correlated winners	stress invariant and dynamic open-interest caps; crowding fee; capacity control	set by the choice of stress moves
Unknown simulator leak	skill-ratio monitoring; challenge window for large claims	detection lag
Contract defects	audits; invariant and differential testing; formal verification of value-moving logic; caps; bug bounty	residual, insured
Governance capture	immutable class; hard bounds; timelock; identity-bound, non-transferable votes	limited by hard bounds

Oracle dependencies. Since Pyth's August 2026 upgrade, a primary report is valid when signed by three of five independently operated routers. Both Pyth and Chainlink now require API credentials to obtain reports, which makes their distribution services a common point of failure. During an outage, market orders settle late but unchanged, while limit and stop fills and breach proofs are delayed; the resulting slippage is left to the insurance fund.

Engine design questions the Yellow Paper leaves open

1. **Inadmissible first reports.** No outcome is yet specified for a market order whose certified first report fails the admissibility checks; such an order freezes its account.
2. **Eroded stress coverage.** Price moves can push the reserve below the stress requirement. New risk is then blocked, but nothing actively restores coverage.
3. **Erasure of breaches.** A breach seen at some report can be erased by a later settlement before its proof lands; the pass and payout bounds assume that monitoring keepers win this race.

4. **Conditional orders.** The no-free-edge result for limit and stop orders is rigorous only within a specific price model, with one remaining technical gap.

Further research problems: keeper rewards that guarantee liveness under congestion; pricing that degrades gracefully in trending markets; optimal joint use of price, menu and capacity under adverse selection; commit privacy (for example encrypted mempools) to reduce free-riding on the copy vault; and cross-chain execution for the copy vault with bounded latency.

Implementation. Value-moving logic is concentrated in three contract modules (trading engine, fee router, vaults) to keep the audit surface small, and is targeted for formal verification before mainnet. Session keys can only place and cancel orders, never move USDC, claim or change the payout address; trader gas is sponsored, so traders never need to hold S.

Reference parameters. The Yellow Paper proposes, for the first deployment and subject to calibration: crypto markets BTC, ETH, S and SOL trading 24/7; a 2 s commit delay and freshness window; 0.5% oracle agreement; a base half-spread of at least 2 basis points; a 5× leverage cap; an 80% profit split; one-day epochs; a coverage target of 1.5 with an upper band of 2.5; a target loss ratio of 0.6; a price rate limit of 5% per day; stress moves of 15% (BTC, ETH) and 25% (S, SOL); and a stress buffer of 0.5.

13 Regulatory considerations

This section raises questions for counsel; it is not legal advice. Depending on the jurisdiction, paid evaluations with monetary rewards may be analyzed as games of chance, as financial instruments or derivatives (in particular contracts for difference), as investment services, or as crypto-asset services. In the EU, the Markets in Crypto-Assets Regulation (MiCA) is relevant to the last of these. Its exclusion for services provided "in a fully decentralised manner without any intermediary" should not be assumed to apply to a protocol in which a multisig proposes parameter changes at launch and a guardian can pause activity.

Fairgame's design is compatible with a conservative posture but does not settle these questions. svUSD is non-transferable and non-redeemable, payouts are performance rewards computed by a published formula, evaluations are priced on a skill-neutral basis, and the protocol never takes custody of trader funds beyond the evaluation fee. Decentralized governance does not by itself shield contributors from liability (see *CFTC v. Ooki DAO*). Before mainnet, the project should obtain jurisdiction-specific opinions, define the legal wrapper of the treasury and its contributors, decide on access restrictions, and screen payout addresses against sanctions lists.

14 Roadmap

The Yellow Paper defines the roadmap as a sequence of phases; it gives **no target dates**.

Phase 0. Research. Build the reference simulator; calibrate rule sets, spreads and controller gains against historical prices and synthetic trader populations; review the Yellow Paper.

Phase 1. Testnet. Deploy the core contracts (excluding later-phase modules) together with keepers, the indexer, the trading interface and the firm-health page; begin audits.

Phase 2. Guarded mainnet. Launch tournaments; then open a small menu of tight rule sets with hard caps on funded capacity; register for Fee Monetization and apply for ecosystem grants.

Phase 3. Self-balancing. Switch on the controller; introduce optional proof of personhood and verified discounts, the passport, scaling and direct funding, and the referral program.

Phase 4. Autonomy. Transition to identity-bound reputation governance; launch the copy-capital vault and net-book hedging.

15 Conclusion

A permissionless prop firm cannot rely on the levers conventional firms use to stay solvent, so Fairgame replaces them with a theory and a control loop. Fair execution caps what any strategy without an edge, or any coalition of them, can extract; those caps set prices that need no identity checks and liabilities that can be published at every block; and a controller with a known stability region steers the rest. The result is a firm whose solvency argument is public and whose rules are the same for everyone, built to find and pay real trading skill.